

## KENOSHA AP, WI, USA

WMO: 726505

Lat: 42.595N

Lon: 87.938W

Elev: 227

StdP: 98.63

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 04845

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 1             | -19.6      | -17.0 | -24.7                         | 0.4 | -19.1 | -22.2 | 0.5 | -16.3 | 12.1                  | -3.5 | 11.1 | -3.0 | 4.4                      | 280  | 0.574 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 32.3            | 23.6 | 30.6 | 22.8 | 28.9 | 21.9 | 25.0                | 30.5 | 24.0 | 28.7 | 23.0 | 27.4 | 5.5                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.1                            | 18.4 | 27.9 | 22.4 | 17.6 | 27.0 | 21.6 | 16.7 | 25.9 | 77.2          | 30.5 | 73.0 | 28.8 | 69.3 | 27.5 | 29.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 11.1              | 9.6  | 8.5 | -23.7                      | 35.1 | 3.4                | 2.3 | -26.2                                              | 36.7 | -28.2      | 38.1 | -30.1      | 39.4 | -32.6      | 41.1 |    |    |
|                   |      |     | -24.0                      | 26.6 | 3.3                | 1.2 | -26.4                                              | 27.5 | -28.4      | 28.2 | -30.2      | 28.9 | -32.7      | 29.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 9.1    | -5.1  | -3.5  | 1.9   | 8.0   | 13.9  | 19.2  | 22.1  | 21.2  | 17.6  | 10.9  | 4.2   | -1.8 |
|                                                                                  | DBStd       | 10.74  | 6.50  | 6.25  | 6.10  | 4.89  | 4.87  | 4.15  | 3.51  | 2.98  | 4.33  | 4.88  | 5.34  | 5.99 |
|                                                                                  | HDD10.0     | 1823   | 469   | 378   | 263   | 94    | 16    | 1     | 0     | 0     | 1     | 48    | 186   | 368  |
|                                                                                  | HDD18.3     | 3718   | 728   | 610   | 511   | 313   | 154   | 38    | 7     | 7     | 66    | 235   | 425   | 625  |
|                                                                                  | CDD10.0     | 1503   | 0     | 1     | 12    | 34    | 138   | 278   | 376   | 348   | 228   | 77    | 11    | 1    |
|                                                                                  | CDD18.3     | 355    | 0     | 0     | 1     | 2     | 18    | 66    | 124   | 97    | 43    | 5     | 0     | 0    |
|                                                                                  | CDH23.3     | 3191   | 0     | 0     | 6     | 37    | 189   | 617   | 1181  | 766   | 354   | 40    | 0     | 0    |
|                                                                                  | CDH26.7     | 985    | 0     | 0     | 1     | 7     | 46    | 195   | 413   | 218   | 101   | 6     | 0     | 0    |
| Wind                                                                             | WSAvg       | 4.1    | 4.8   | 4.7   | 4.6   | 4.9   | 4.3   | 3.6   | 3.3   | 3.1   | 3.4   | 4.1   | 4.5   | 4.6  |
| Precipitation                                                                    | PrecAvg     | 878    | 44    | 38    | 55    | 95    | 97    | 104   | 84    | 94    | 81    | 70    | 65    | 49   |
|                                                                                  | PrecMax     | 1056   | 94    | 90    | 111   | 181   | 268   | 217   | 185   | 285   | 176   | 164   | 136   | 103  |
|                                                                                  | PrecMin     | 610    | 9     | 4     | 15    | 46    | 16    | 17    | 42    | 36    | 17    | 24    | 15    | 16   |
|                                                                                  | PrecStd     | 116    | 23    | 24    | 30    | 38    | 53    | 50    | 33    | 48    | 39    | 39    | 35    | 24   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 12.3  | 16.0  | 23.0  | 27.5  | 30.5  | 33.5  | 35.9  | 33.6  | 32.0  | 27.4  | 20.3  | 15.3 |
|                                                                                  |             | MCWB   | 9.7   | 11.6  | 16.7  | 17.2  | 20.9  | 23.4  | 25.2  | 24.2  | 22.8  | 20.3  | 14.7  | 12.9 |
|                                                                                  | 2%          | DB     | 8.0   | 10.9  | 18.0  | 23.0  | 27.9  | 31.1  | 32.5  | 31.1  | 29.5  | 23.9  | 17.3  | 12.0 |
|                                                                                  |             | MCWB   | 5.3   | 7.6   | 13.4  | 15.3  | 19.6  | 22.5  | 24.4  | 23.4  | 21.9  | 17.9  | 12.4  | 9.9  |
|                                                                                  | 5%          | DB     | 5.0   | 7.2   | 14.7  | 20.0  | 25.6  | 28.9  | 30.8  | 28.9  | 27.4  | 21.5  | 14.8  | 8.2  |
|                                                                                  |             | MCWB   | 3.1   | 4.7   | 10.7  | 13.6  | 18.6  | 21.8  | 23.3  | 22.3  | 20.3  | 16.5  | 11.1  | 6.3  |
|                                                                                  | 10%         | DB     | 3.0   | 4.1   | 11.0  | 17.0  | 22.8  | 27.1  | 28.8  | 27.4  | 25.1  | 18.9  | 12.4  | 6.2  |
|                                                                                  |             | MCWB   | 1.8   | 2.3   | 7.8   | 11.4  | 17.0  | 20.7  | 22.2  | 21.3  | 19.6  | 14.8  | 9.1   | 4.2  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 10.7  | 12.3  | 17.2  | 18.9  | 22.8  | 25.2  | 26.9  | 25.5  | 24.6  | 22.0  | 15.7  | 13.0 |
|                                                                                  |             | MCDB   | 11.7  | 13.9  | 22.9  | 25.7  | 28.3  | 30.9  | 32.3  | 31.5  | 29.5  | 25.9  | 18.7  | 14.5 |
|                                                                                  | 2%          | WB     | 5.6   | 7.8   | 14.3  | 16.3  | 20.8  | 23.8  | 25.5  | 24.3  | 22.9  | 19.2  | 13.7  | 10.0 |
|                                                                                  |             | MCDB   | 7.6   | 9.9   | 17.5  | 21.9  | 26.2  | 29.1  | 31.0  | 29.1  | 27.1  | 22.2  | 16.6  | 11.3 |
|                                                                                  | 5%          | WB     | 3.4   | 4.7   | 11.1  | 14.3  | 19.5  | 22.6  | 24.3  | 23.4  | 21.8  | 17.3  | 11.5  | 6.7  |
|                                                                                  |             | MCDB   | 4.8   | 7.1   | 13.9  | 19.0  | 24.0  | 27.6  | 29.3  | 27.5  | 25.7  | 20.1  | 13.9  | 8.2  |
|                                                                                  | 10%         | WB     | 1.9   | 2.7   | 7.7   | 11.7  | 17.9  | 21.5  | 23.3  | 22.4  | 20.5  | 15.5  | 9.3   | 4.2  |
|                                                                                  |             | MCDB   | 3.1   | 4.1   | 10.6  | 16.2  | 22.1  | 25.9  | 27.6  | 26.0  | 24.1  | 18.6  | 12.3  | 5.9  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.6   | 8.2   | 9.0   | 10.4  | 11.1  | 10.9  | 11.1  | 10.7  | 11.3  | 10.0  | 8.5   | 7.3  |
|                                                                                  |             | MCDBR  | 9.1   | 10.8  | 13.4  | 15.9  | 14.1  | 12.8  | 12.2  | 12.5  | 13.2  | 13.2  | 11.8  | 9.3  |
|                                                                                  |             | MCWBR  | 7.5   | 8.2   | 9.6   | 9.4   | 8.0   | 6.5   | 5.7   | 6.0   | 6.6   | 7.9   | 8.4   | 7.7  |
|                                                                                  | 5% WB       | MCDBR  | 8.5   | 10.4  | 12.2  | 14.5  | 12.7  | 11.4  | 10.9  | 10.5  | 11.4  | 11.2  | 10.6  | 8.5  |
|                                                                                  |             | MCWBR  | 7.3   | 8.3   | 9.5   | 9.7   | 8.0   | 6.6   | 5.9   | 5.7   | 6.4   | 8.1   | 8.5   | 7.7  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.290  | 0.309 | 0.341 | 0.392 | 0.416 | 0.435 | 0.433 | 0.425 | 0.395 | 0.348 | 0.310 | 0.289 |      |
|                                                                                  | taud        | 2.430  | 2.382 | 2.393 | 2.279 | 2.262 | 2.257 | 2.263 | 2.284 | 2.341 | 2.472 | 2.519 | 2.474 |      |
|                                                                                  | Ebn at Noon | 855    | 892   | 902   | 875   | 860   | 842   | 839   | 834   | 833   | 838   | 829   | 826   |      |
|                                                                                  | Edh at Noon | 76     | 94    | 105   | 127   | 133   | 134   | 132   | 126   | 110   | 86    | 70    | 67    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.49   | 2.40  | 3.56  | 4.60  | 5.29  | 6.05  | 6.13  | 5.31  | 4.32  | 2.79  | 1.74  | 1.23  |      |
|                                                                                  | RadStd      | 0.20   | 0.28  | 0.31  | 0.38  | 0.41  | 0.40  | 0.28  | 0.33  | 0.36  | 0.32  | 0.17  | 0.13  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (61 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page